Impact of FinTech on Stock Price Liquidity

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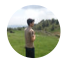

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Data can be obtained from the sources described in the study.

Conflict of interest
Authors declare no conflict of interest.

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Abstract

We examine whether and how financial technology (FinTech) influences stock price liquidity. FinTech was measured using the digital finance index from the Institute of Digital Finance of Peking University (PKU-digital finance IIC), the Shanghai Finance Institute, and Ant Financial Services Company. By using a sample of Chinese listed firms from 2011 to 2022, results from ordinary least square regression indicate that Fintech is positively associated with stock price liquidity. The relationship between FinTech and stock price liquidity is more pronounced in the presence of higher media coverage, analysts' following, and in non-state-owned enterprises but weaker when economic policy uncertainty is higher. The findings remain consistent with alternative measures of FinTech, omitted variables problems, and endogeneity issues. Results provide insights that the development of FinTech increases financial transparency, which promotes the information environment, thereby increasing stock liquidity. Findings contribute to the literature on the governance role of FinTech in influencing stock liquidity.

Keywords: FinTech; stock price liquidity; information asymmetry; China.

1. Introduction

The development of FinTech (or digital finance)[1], such as big data, crowdfunding, peer-to-peer (P2P) lending, cloud computing, and blockchain, has gained enormous attention around the world, especially in emerging economies. FinTech is a phenomenon that integrates digital technologies and finance that plays a significant role in emerging economies. It not only improves the efficiency of financial services with comprehensible advanced technological platforms but also intensifies the breadth and depth of financial services. FinTech offers an extensive range of services (financial), products, software, and unique communication channels (Gomber et al., 2017). The development of FinTech facilitates digital innovation, enhances traditional financial systems, and curbs information asymmetry. It may shorten the loan processing time, thereby improving the efficiency of financial institutions' services (Fuster et al., 2019). The majority of the studies have explored the influence of FinTech on the financial sector (Buchak et al., 2018; Cheng & Qu, 2020; Daud et al., 2022; Wang et al., 2020; Zhao et al., 2022) and macro-economy (Aziz & Naima, 2021; Tang et al., 2019; Zhang et al., 2019), with only a few examining how it influences firms from a micro perspective. In this study, we seek to fill the gap in the literature by exploring how FinTech influences the liquidity of stock prices.

Stock price liquidity has received considerable importance from regulators, investors, and academics, especially after the 2008 financial crisis. Handa and Schwartz (1996) reveal that stock liquidity is a critical market feature. Investors usually expect higher returns when they invest in less liquid assets (Amihud & Mendelson, 1986), which consequently increases the cost of equity (Butler et al., 2005). The impact of firm performance (Fang et al., 2009) provokes firms to endeavor for higher liquidity (Levine & Zervos, 1996). Research has demonstrated that

[1] FinTech and digital finance are used interchangeably in literature.

information transparency is an important factor that may affect the liquidity of stocks (e.g., Boubaker et al., 2019; Diamond & Verrecchia, 1991; Majeed & Yan, 2022). Prior literature also documents that disclosure quality and financial reporting are linked with stock liquidity (Frino et al., 2013; Welker, 1995). Studies argue that increased information transparency can mitigate adverse selection, which results in higher stock liquidity (Ascioglu et al., 2012). Although there is extensive research on the determinants of stock liquidity, there is no evidence on whether it is influenced by FinTech.

We argue that the development of FinTech is positively associated with the liquidity of stock prices for two reasons. First, using emerging technologies (e.g., machine learning and big data) may help to secure accurate information related to firms' operations (Buchak et al., 2018). Hence, FinTech not only increases the quality of business and accounting operations but also increases access to finance resulting in a lower opaque information environment, thus, increasing stock liquidity. Second, the development of FinTech mitigates agency problems related to information asymmetry and acts as an external governance mechanism (Ji et al., 2022; Zhu, 2019). Extant studies documented that improved governance leads to higher stock liquidity (Ali et al., 2017). FinTech (such as big data) increases the availability of timely information which ultimately curbs agency conflicts (Zhu, 2019). Therefore, we argue that improved governance from FinTech can lead to higher stock price liquidity.

Using a sample of Chinese A-share listed firms from 2011 to 2022 and considering the FinTech index developed by The Peking University Digital Financial Inclusion Index of China, we find robust evidence that FinTech is positively associated with stock liquidity. In our additional analyses, we show that the positive effect of FinTech on stock liquidity is stronger for firms with higher media coverage and higher analyst following, but weaker for firms owned by the state and in the presence of higher uncertainty. The results remain consistent with alternative proxies and after addressing issues related to omitted variables and endogeneity.

This study offers two important contributions. First, this study contributes to the literature on FinTech. Although studies have explored the effect of FinTech on the financial sector, such as total factor productivity (Shen & Guo, 2015), bank credit risk (Cheng & Qu, 2020), shadow banking (Buchak et al., 2018), financial efficiency (Wang et al., 2020), financial stability (Daud et al., 2022), and bank performance (Zhao et al., 2022), only a few have examined its influence at the firm level, corporate bankruptcy risk (Ji et al., 2022), investment efficiency (Lv & Xiong, 2022), and cash holdings (Shao et al., 2022), excess leverage (Lai et al., 2023), corporate risk-taking (Tang et al., 2024), and cross-regional investment (Huang et al., 2024). Our study contributes to the literature by providing new evidence on how the liquidity of stock prices can be influenced by the development of FinTech. Unlike the existing literature, we do not focus on the firm's decision but rather we focus on investors' decision, by investigating how FinTech increased stock price liquidity.

Second, this study contributes to stock liquidity literature. Although the literature has explored several factors influencing stock liquidity (e.g., Boubaker et al., 2019; Chung et al., 2010; Frino et al., 2013; Pham, 2020; Zhang et al., 2020), there is no evidence on how FinTech can influence stock liquidity. To the best of our knowledge, this study provides novel insights that the development of FinTech may significantly affect the liquidity of stock prices.

The rest of this paper is organized as follows. Section 2 presents the literature and develops the hypothesis. Section 3 describes methods. Section 4 presents the main results and robustness checks. Section 5 discusses the findings from additional analyses. The final section concludes.

2. Literature review and hypothesis

The development of FinTech introduced new tools, techniques, and solutions to lending equity markets that have substantially influenced firms' access to funds. This property is called financial inclusion. FinTech has significantly reshaped the way through which financial services are delivered by firms. The prime purpose of FinTech is its credibility and strength of improving financial inclusion. Considering the importance of FinTech, studies have examined and shown that FinTech has improved the accessibility of farmers to capital resources (Masakara, 2020), alleviated the cost of borrowing (de Roure et al., 2021), enhanced the access of SMEs (Sheng & Fan, 2020), female entrepreneur (Mollick & Robb, 2016), and younger top management team (Cumming et al., 2021) to formal financial resources.

The growing importance of FinTech and its potential benefits at the macro-level have led scholars to explore the effect of FinTech at micro-level (firm-level). For example, Shao et al. (2022) find that the development of FinTech is negatively associated with the financial constraint of non-state-owned firms in emerging markets. Cao et al. (2021) argue that FinTech significantly improves the energy-environmental performance in China. Chen and Zhang (2021) document that the servitization of the manufacturing industry is significantly improved with FinTech. Tang et al. (2022) show that the value of strategic emerging firms is significantly improved with the development of FinTech. Lv and Xiong (2022) argue that FinTech positively affects investment efficiency. Ji et al. (2022) reveal that FinTech decreases the bankruptcy risk of firms. In this study, we extend the literature on how the development of FinTech influences the liquidity of stock prices. Lai et al. (2023) argue that excess leverage becomes lower with FinTech. Tang et al. (2024) show that FinTech increases corporate risk-taking. Huang et al. (2024) suggests that FinTech can promote cross-regional investment.

We carry investigation considering the Chinese context for several reasons. First, China has been at the forefront of the development of FinTech. According to a Briefing report, China has been the global leading FinTech investor since 2018. FinTech investment in China was US$ 25.5bn[2] in 2018. This was a significant increase (approximately over 900%) in FinTech, yielding over 50% of the global FinTech investment. In 2019, China's FinTech sector had a market value of US$59.2bn, and it is predicted that the market size will be US$ 85.7 by 2022. Chinese financial sector is less developed compared to developed countries which creates more opportunities for innovation (Goldstein et al., 2019).

Second, China is the second largest, but still an emerging economy. It has a unique business, institutional, and legal environment, which is distinct because of the accounting information environment, finance, and governance structure (Allen et al., 2012; Conyon & He, 2011; Jiang & Kim, 2015). However, the Chinese market is still underdeveloped (Jiang & Kim, 2020). The Chinese financial market is characterized by poor informational transparency because the political, legal, and cultural environment disturbs the transformation of unbiased and timely financial information, leading to higher opacity. Therefore, exploring the factors determining stock liquidity in the Chinese context may provide valuable insights.

Finally, it is difficult to measure FinTech. However, scholars have developed a more reliable method to measure FinTech in the Chinese context. The Ant Financial, the Institute of FinTech at Peking University created the "Digital Financial Inclusion Index" (PKU-DFIIC), which measures a FinTech development index of all provinces and cities across the country. This index has been developed since 2011.

Stock liquidity reflects the ability to quickly trade large quantities of security with minimal trading cost and lower price impact. Liu (2006) refers to liquidity of a stock if its large volumes

[2] https://www.china-briefing.com/

may be traded with little or no price impact. Research (e.g., Amihud, 2002; Amihud & Mendelson, 1986; Hasbrouck, 2006; Le & Gregoriou, 2020) have explained stock liquidity from four different dimensions: trading speed (how quickly can a stock be traded at a particular cost with given quantity), trading quantity (how much security can be traded at a given cost), price impact (how easy it is to trade the security of a given quantity with minimum impact on price), and trading costs (all expenses related to the trade of a given quantity of an asset).

Many studies have examined the factors that influence stock liquidity. Studies have elaborated that stock liquidity can be influenced by investors' characteristics such as investors' sentiments and investment horizons (Kumari 2019; Wang & Wei 2021). Studies have described that stock liquidity can be influenced by the characteristics of firms, boards, and directors, such as employee ownership, cultural diversity in ownership, top management council, and CEOs with law degrees (Jung & Choi, 2021; Michael et al., 2022; Pham, 2020; Zhang et al., 2020), among other. Despite the extensive research (e.g., Aghanya et al., 2020; Dakse et al., 2008; Farino et al., 2013; Heflin et al., 2005; Michael et al., 2022; Pham, 2020; Wang & Wei, 2021), there is no evidence on whether and how FinTech influences stock price liquidity.

We argue that the development of FinTech improves the liquidity of stocks by enhancing information transparency. Several studies explained the role of financial disclosure in ameliorating stock price liquidity. For example, Diamond and Verrecchia (1991) document that higher financial disclosure is a channel through which the adverse selection issue is attenuated, resulting in higher stock liquidity. Heflin et al. (2005) argue that greater disclosure quality promotes an information environment that leads to higher stock liquidity (i.e., low spread). Dakse et al. (2008) suggest that IFRS enhances the transparency of financial reporting, further increasing liquidity. Farino et al. (2013) document that IFRS improves financial transparency, curbs information asymmetry, and lead to higher liquidity. Therefore, we believe that stock liquidity can increase with the development of FinTech, because FinTech may improve firms' information environment.

Prior studies (e.g., Ji et al., 2022; Lv & Xiong, 2022; Zhu, 2019) described how information transparency can be increased with the development of FinTech development. Information transparency decreases when some participants in the market have more private information in the market compared to others. This information asymmetry among market participants (informed and uninformed participants) creates market friction leading to the problem of adverse selection. Adverse selection attenuates the trading of investors (uninformed) since informed investors have private information. Consequently, this leads to lower stock liquidity due to lower demands for stock (Copeland & Galai, 1983). When firms disclose more information, the information environment is improved, and therefore investors are well informed (Diamond, 1985). Moreover, increased public information curbs the private information collection incentive, which lowers profits from privately informed trading (Diamond, 1985).

The development of FinTech plays an important role in mitigating the information asymmetry of firms (Demertzis et al., 2018; Shao et al., 2022). FinTech increases the flow of information and reduces the cost of financial information processing. Lower information asymmetry arising from the development of FinTech helps firms to monitor the opportunistic behavior of the firms leading to higher investment efficiency (Lv & Xiong, 2022), reduced corporate risk (Ji et al., 2022), enhanced firm value (Tang et al., 2022), and improved manufacturing servitization (Chen & Zhang, 2021). This suggests that the development of FinTech increases the flow and transparency of information, thereby decreasing information asymmetry.

Based on the aforementioned discussion, we assume that the development of FinTech increases the availability of information, which reduces information asymmetry. Further, with improved transparency and lower information asymmetry, stock liquidity can be significantly increased. Thus, we expect that the development of FinTech can increase information transparency and transaction costs, which leads to higher stock liquidity. Based on these arguments, we hypothesize that:

Hypothesis 1. *The development of FinTech is positively associated with stock price liquidity.*

3. Methods

3.1 Data and sample

Our sample comprises all A-share Chinese firms listed on Shenzhen and Shanghai stock exchanges during the period 2011 to 2022. The sample period starts from 2011 because the FinTech index was developed this year. All the data is obtained from the China Stock Market and Accounting Research (CSMAR) database. Our sample initially consisted of 30,852 firm-year observations from 4,937 listed companies. We select the final sample using the following criteria. First, we excluded firms in the financial industry (2,403 observations) because of the unique industry features and different regulatory and disclosure requirements. Second, we excluded firms that are labeled as ST (841 observations) because they are under strict regulatory scrutiny. Finally, we excluded missing data on variables (6,954 observations). Using these criterions, we obtained a sample comprising 20,654 firm-year observations. Figure 1 provides flow chart of methodology.

[Insert Figure 1 about here]

3.2 Measurement of stock price liquidity

We used four proxies for stock price liquidity. The first measure is bid-ask spread (denoted by *Spread*) computed based on daily low/high stock prices (e.g., Corwin & Schultz, 2012). This measure is one of the reliable methods for stock liquidity measurement and has been used extensively in the literature (Dang et al., 2022; Jung and Choi, 2021; Roy et al., 2022). This method estimates the spread as a function of high–low ratios from the first- and second-day intervals. The following equations (1) and (2) are then estimated:

$$\delta_{kt} = \sum_{a=0}^{1}\left[\ln\left(\frac{H^{o}_{kt,\,d+j}}{L^{o}_{kt,\,d+j}}\right)\right]^{2} \tag{1}$$

$$\phi_{kt} = \sum_{a=0}^{1}\left[\ln\left(\frac{H^{o}_{kt,\,d+j}}{L^{o}_{kt,\,d+j}}\right)\right]^{2} \tag{2}$$

In equations (1) and (2), L^{O} is low price and H^{O} indicates high price. Following the literature on high-low price ratios, Corwin and Schultz (2012) achieve the following equations (3) and (4) for the spread (*Spread*):

$$Spread_{kt} = \frac{2(e^{\mu_{it}} - 1)}{1 + e^{\mu_{it}}} \tag{3}$$

$$\text{where } \mu_{kt} = \frac{\sqrt{2\beta_{kt}} - \sqrt{\beta_{kt}}}{3 - 2\sqrt{2}} - \sqrt{\frac{\alpha_{kt}}{3 - 2\sqrt{2}}} \tag{4}$$

We computed the high-low spread for each two-day interval by considering daily high-low prices. Stock's bid-ask spread was computed for each month by averaging all spreads across all two-day intervals. All negative estimations are replaced with zero to calculate monthly average.

Second, we used a stock liquidity proxy (denoted by *Amihud*) developed by Amihud (2002). This method measures the sensitivity of stock price to RMB trading volume. This measure has been used extensively by prior studies (Alp et al., 2021; Chatterjee et al., 2021; Dang et al., 2022;

Pham, 2020; Zhang and Lence, 2022). We estimated Amihud illiquidity measure using daily stock returns, the number of trading days, and trading volume in the Chinese yuan.

$$Amihud_{ky} = \frac{1}{D_{ky}} \sum_{d=1}^{D_{kt}} \frac{|R_{ktd}|}{VOL_{kyd}} \tag{5}$$

Equation (5) is followed for estimating illiquidity ratio firm "*k*" in year "*y*" with return "*R*" for each day "*d*". *"VOL"* is volume of stock "*i*" stock day *"d"*. "*D*" is number of zero trading days for which data is available. A higher value of the Amihud measure signifies lower stock liquidity.

Third, following Corwin and Schultz (2012), all negative estimations are replaced with 0 for calculation of monthly average for the third measure of stock liquidity (denoted by *STO*). This method has been widely used in literature (Fengrong et al., 2022; Kumari, 2019; Wang and Wei, 2021). Stock turnover (*STO*) is trading frequency: the sum of total (daily) shares traded in the market for each year scaled by the outstanding shares.

$$STO_{it} = \frac{VOL_{it}}{N_{it}} \tag{6}$$

In equation (6), VOL_{it} represents shares traded and N_{it} is outstanding shares. Higher values of *STO* indicate higher stock liquidity.

Our fourth measure of stock liquidity is the trading volume (*Volume*) which is the natural log of the total number of shares traded on the stock exchange (Ali et al., 2017). This method has been used in literature (Wang and Wei, 2021). Stock liquidity is higher with higher volume.

3.3 Measurement of FinTech

FinTech is measured using the digital finance index from the Institute of Digital Finance of Peking University (PKU-digital finance IIC), the Shanghai Finance Institute, and Ant Financial

Services Company. This index covers three main dimensions of digital finance services: depth of use (DU), breath of coverage (BC), and digital support services. The overall index is composed of six subcomponents: investment, insurance, credit, monetary funds, credit investigation, and payment. In baseline regression, to measure FinTech, we use three variables i.e., overall index (*FINTECH*), breath of coverage index (*FINTECH_BC*), and depth of use index (*FINTECH_DU*) at provincial level. We normalize FinTech index following literature (e.g., Ji et al., 2022). This index has been used by several studies (e.g., Ji et al., 2022; Shao et al., 2022; Tang et al., 2022).

3.4 Model

We use the following OLS regression to examine the effect of FinTech on stock liquidity.

$$LIQUIDITY_{it} = \beta_0 + \beta_1 FinTech_{it} + \beta_2 MTB_{it} + \beta_3 SIZE_{it} + \beta_4 R\&D_{it} + \beta_5 CASH_{it} + \beta_6 RETVOL_{it} + \beta_7 AGE_{it} + \beta_8 TANG_{it} + \beta_9 INVP_{it} + \beta_{10} LEV_{it} + \beta_{11} SOE_{it} + Year + Industry + \sigma_{it}, \qquad (7)$$

In equation (7), *i* represents firm and *t* represents year; parameters β_1 to β_{11} are the corresponding coefficient of variables, and σ_{it} is the error term. $LIQUIDITY_{it}$ represents measures of stock price liquidity (*Spread*, *Amihud*, *STO*, and *Volume*) for firm i in year t. $FinTech_{it}$ indicates FinTech indexes that include overall digital index (*FINTECH*), breadth of coverage (*FINTECH_BC*), and depth of usage (*FINTECH_DU*) for firm *i* in year t. Following the literature, we controlled for several variables in the model. We include the market-to-book ratio (*MTB*), measured as the market value of equity divided by the book value of equity. We controlled for firm size (*SIZE*), measured as the natural log of total assets (Pham, 2020; Roy, Rao, & Zhu, 2022). We also control for research and development intensity (*R&D*) measured as R&D divided by total assets since R&D payoffs are challenging to estimate, leading to lower liquidity. We control for cash holding (*CASH*), calculated as the sum of year-end cash and short-term securities scaled by total sales (Roy et al., 2022). We used stock return volatility (*RETV*) measured as the annual average daily stock return volatility (Roy et al., 2022). We also control firm age (*AGE*) as investors prefer older firms (Atawnah et al., 2018). We include tangibility (*TANG*) measured as total fixed assets divided by total assets. We controlled for the inverse of the stock price (*INVP*) because research shows that low-price firms face higher spreads (Welker, 1995). The literature argues that firms' capital structure selections could impact stock liquidity as firms with higher information asymmetry depend more on debt capital (Myers & Majluf, 1984). We included leverage (*LEV*), computed as the debt-to-equity ratio (Roy et al., 2022). We also controlled for state ownership (*SOE*) using a dummy variable that holds the value of one if the firm is state-owned, otherwise zero. Finally, we controlled for year and industry. Table A1 in Appendix provides detailed explanations of variables.

3.5 Summary statistics and correlation analysis

Table 1 presents descriptive statistics for the main variables. The mean value of the spread (bid-ask) is -0.156 and its standard deviation (SD) is -0.160. We observe a higher SD of liquidity variables, which suggests higher volatility in the stock's liquidity. The summary statistics of the stock liquidity measures are close to those presented in prior studies (e.g., Majeed & Yan, 2022; Trinh et al., 2021). The SD of *FINTECH*, *FINTECH_BC*, and *FINTECH_DU* are 82.031, 90.093, and 72.914, respectively. Thus, the volatility of FinTech development in our sample is considerable. However, the variance is significant, indicating a gap in FinTech development among different regions in China. The summary statistics for the control variables are consistent with the literature.

Table 2 shows the results of the Pearson correlation matrix. Consistent with expectations, there is a significant correlation between the measures of FinTech and stock price liquidity. The results also suggest significant correlations between control variables and FinTech.

[Insert Tables 1 and 2 here]

4. Results

4.1 Baseline results

Table 3 presents the main findings. The results show that coefficients on all proxies of FinTech (*FINTECH*, *FINTECH_BC*, and *FINTECH_DU*) are statistically and significantly negative in columns (1) to (6), whereas statistically positive in columns (7) to (12). The results strongly support our hypothesis that FinTech is positively associated with stock price liquidity. As for economic significance, the coefficient on FINTECH in column (1) of Table (3) suggests that one standard deviation increase in FinTech will increase stock liquidity by approximately 10.25%.

The results for control variables are consistent with the literature. Findings show that firms with large size (*SIZE*), higher return volatility (*RETV*), inverse price (*INVP*), and market-to-book ratio (*MTB*) have higher stock liquidity. Similarly, we find that state-owned firms (*SOE*) have higher liquidity. On the other hand, results indicate that higher leverage (*LEV*), tangibility (*TANG*), and firm age (*AGE*) decrease stock price liquidity. Moreover, we find that research and development (*R&D*) and cash holding (*CASH*) have no significant effect on stock price liquidity.

[Insert Table 3 here]

4.2 Robustness checks

4.2.1 Alternative proxies

We check the robustness of findings using alternative measures. First, we used the Amivest liquidity ratio (*Amivest*), as an alternative proxy for stock price liquidity. This measure has been widely used in prior research (Ali et al., 2017; Roy et al., 2022; Fengrong et al., 2022). We measured Amivest as follow.

$$AMIVEST_{l,i} = \frac{1}{D_{i,m}} \sum_{t=1}^{N_{i,m}} \frac{VOL_{i,t,m}}{\left|R_{i,t,m}\right|} , \tag{8}$$

In equation (8), "*VOL*" represents the volume of shares traded daily, and *R* represents absolute daily stock returns. Results reported in Table 4 using this additional measure of stock liquidity (*Amivest*) remain consistent.

[Insert Table 4 here]

Second, following the literature (e.g., Cao et al., 2021; Fonseka et al., 2021; Wang et al., 2022) we use an alternative proxy of FinTech considering the digitization index (denoted as *FINTECH_DL*), which reflects the trading costs and efficiency of the area by measuring the percentage of mobile payment, loan rates for small and micro firms and popularity of QR code scan. Results using this measure reported in Table 5 remain consistent.

[Insert Table 5 here]

Third, following Ji et al. (2022), we replace the main DF measures, i.e., FINTECH, FINTECH_BC, and FINTECH_DU (provincial level), with city level. FINTECH_City is the overall DF index measured at the municipal level, and FINTECH_B_City, and FINTECH_D_City are the coverage breath index and depth of usage index at the municipality level, respectively. The results reported in Table 6 are consistent with the main regression results.

[Insert Table 6 here]

4.2.2 Omitted variable problem

In this section, we carry out additional analyses to address omitted variable problems. First, we employ firm fixed effects to control for time-invariant attributes, which could be linked to both FinTech and stock liquidity. Results are presented in Table 7, which remain consistent.

[Insert Table 7 here]

Second, unobservable variables may cause a spurious correlation between FinTech and stock price liquidity. To deal with this issue, following the literature (e.g., Moser & Voena, 2012), we included interaction terms of year and industry. The results reported in Table 8 remain consistent.

[Insert Table 8 here]

4.2.3 Endogeneity issues

Although we find that FinTech is positively associated with stock price liquidity, this association can suffer from endogeneity issues. To deal with this issue, we use the two-stage least-square (2SLS) method. The first step in this approach is to find a valid instrument. Following Ji et al. (2022) and Tang et al. (2020), we use internet penetration (IP) as an instrument variable, computed using Internet Development Index (IDI) of the firms registered in a province[3]. We use this as an instrument variable for several reasons. First, the internet penetration in a province impacts the development of FinTech in that province. Second, the level of internet penetration doesn't have a direct influence on stock price liquidity but is directly associated with FinTech. Third, this instrument variable passes the under-identification diagnostic test, as *Kleibergen-Paap rk LM* statistics (statistically significant at 1%) and the weak identification tests, such as the *Crag – Donal Wald F-statistics* and the *Kleibergen-Paap rk Wald F-statistics* (statistically significant at 1%). Thus, internet penetration is a valid instrument.

In first stage regression, FinTech (i.e., *FINTECH*, *FINTECH_BC*, and *FINTECH_DU*) is the dependent variable, whereas internet penetration is the explanatory variable, along with the same control variables in the main equation. The first stage regression results reported in Models (1) to (3) of Table 8 indicate significantly negative coefficients on internet penetration (*IP*), which further confirms the validity of instrument. Most importantly, the results are significant for all measures of FinTech.

[3] Data is obtained from "Chinese Internet Development Statistics".

In second stage, the residuals from the first stage regression are used as a proxy of FinTech to predict stock liquidity. Results in Models (4) to (15) in Table 9 remain consistent. Most important, the results are significant for all measures of stock price liquidity. The results in this section suggest that our findings remain consistent after addressing issues related to endogeneity.

[Insert Table 9 here]

4.3 Additional Analysis

We extend our main findings using several additional analyses. First, we examine the effect of FinTech on stock liquidity for firms with high and low media coverage. Second, we investigate the effect of FinTech on stock price liquidity in the presence of high and low analysts following. Third, we conduct a subsample analysis by dividing the sample into SOEs and Non-SOEs. Finally, we examine whether and how economic policy uncertainty influences the association between FinTech and stock liquidity.

4.3.1 Role of media coverage

Media plays a significant role in monitoring a firm's behavior and building mechanisms through which valuable information is conveyed (Chen et al., 2022). Existing literature shows that media coverage influences corporate decisions (e.g., Dai et al., 2021; Drake et al., 2014; Wang & Zhang, 2021). Research suggests various attributes that influence stock liquidity. It elaborates that any governance tool (external or internal) that enhances the information environment can lead to higher stock liquidity. Therefore, the role of media coverage as external governance can't be ignored regarding its influence on stock liquidity. One of the important roles of the media is to disclose and report events that occur within a firm, which benefits firms by mitigating information asymmetry and opaqueness. Media continuously produce and spread information in the market, which meets stakeholders' requirements for information (Drake et al., 2014; Graf-Vlachy et al., 2020). For instance, Pollock and Rindova (2003) document that media coverage affects the behavior of potential investors represented by a firm's IPO performance. Media provides information that may affect stakeholders' decisions about firms and may also stock liquidity.

We argue that media coverage can influence the association between FinTech and stock liquidity for two reasons. First, media coverage acts as an external governance mechanism that releases public opinion about firms', thereby disposing of early warnings. Firms are required to improve the information environment, thus boosting stock price liquidity. Second, based on the classical asset pricing model, disseminating information would increase future cash flow expectations. The proliferation of information through media can increase the speed of information transmission and improve the capital market's informational efficiency, thus, leading to greater stock liquidity. Based on these two reasons, we expect that the effect of FinTech on stock liquidity is higher with higher media coverage compared with low media coverage because firms with high media coverage have low information opacity and high-quality information environment.

To examine the role of media coverage, we divided our sample into high and low media coverage based on industry median. Following the literature (e.g., Cheng et al., 2022; Liang et al., 2022), we measured media coverage as natural logarithm of total number of firms' news. Results reported in Panel A of Table 10 indicate significantly positive coefficients on all proxies of FinTech, whereas the coefficients are insignificant in Panel B of Table 10. Results are consistent with expectations that the FinTech influence on stock liquidity is greater with high media coverage compared with low media coverage.

[Insert Table 10 here]

4.3.2 Role of analyst

Financial analysts play an intermediary role in the stock market by providing firms with information, including buy-sell recommendations and earnings price forecasts (Healy & Palepu, 2001). We argue that analysts can significantly influence the association between FinTech and stock liquidity. A major reason is that analysts can significantly influence the information environment of firms. Healy and Palepu (2001) argue that information asymmetry is low when more analysts follow firms, because analyst offers a vital contribution to the information environment related to firms (Bushman, 2011). Yu (2010) argued that analysts understand the implications of corporate governance for future cash flows and suggested that higher analysts disclose more information, thereby improving the information environment. Jiraporn et al. (2012) indicate that analysts work as information intermediaries, provide oversight over managers and thus help mitigate agency conflict. Gao et al. (2020) argue that analysts, an essential capital market intermediary, engage in information production and perform the external monitoring role on their covered firms (Irani & Oesch, 2016). Therefore, we expect that the FinTech influence on stock liquidity is stronger for firms with higher analysts compared to firms with low analysts following.

To test the above arguments, we divide our sample into low and high analyst following based on the total number of analysts following (log-transformed). Results in Panel A of Table 11 indicate statistically significant coefficients on all proxies of FinTech, whereas coefficients are statistically insignificant in Panel B. The findings are consistent with expectations by indicating that the FinTech influence on stock liquidity is more pronounced for firms with higher analysts following compared to firms with fewer analysts following.

[Insert Table 11 here]

4.3.3 Subsample analysis using state-owned and non-state-owned firms

We further examine whether the effect of FinTech on stock liquidity varies across state-owned (SOEs) and non-state-owned firms (Non-SOEs). SOEs are very different from privately-owned firms. Unlike profit-oriented private firms, SOEs carry out several social and political goals since they are controlled by the government. In addition, SOEs benefit from a soft budget, because the government can reduce financial constraints by offering implicit guarantees, tax discounts, credit, and other beneficial financial supports.

We argue that the impact of FinTech on stock liquidity is stronger in Non-SOEs than SOEs. The political view posits that the government uses SOEs to pursue a political objective, which rarely coincides with profit maximization (Boycko et al., 1994). Because investors associate state ownership with poor governance (Borisova et al., 2012), they are less likely to buy stocks frequently. Chung et al. (2010) argue that efficient corporate governance leads to high market liquidity, whereas Kyle (1985) argued that trading is more significant for firms with weak corporate governance. However, Boubakri et al. (2020) found a non-monotonic association between stock liquidity and state ownership.

This aforementioned discussion motivates us to examine the effect of FinTech on stock liquidity by dividing the sample into SOEs and Non-SOEs. Results in Panel A of Table 12 indicate insignificant coefficients on all proxies of FinTech in a sample of SOEs, while the coefficients are statistically significant in Panel B for a sample of Non-SOEs. Results support our expectations that FinTech influence on stock liquidity is greater in Non-SOEs compared to SOEs.

[Insert Table 12 here]

4.3.4 Effect of uncertainty

We further examine how uncertainty influences the FinTech and stock liquidity association. Prior studies suggest that uncertainty can significantly influence information asymmetry, thereby influencing stock liquidity. Investors are opposed to uncertain shocks, and their demand for accurate financial information increases when uncertainties rise (Drechsler, 2013). Thus, information asymmetry is more significant to a firm when uncertainty is higher (Nagar et al., 2019). Increase in information asymmetry lowers investors' appetite to trade and negatively affects stock liquidity (Schoenfeld, 2017). Further, under high uncertainty, firms are cautious about information disclosure and carefully decide on what to disclose and what not to. Studies (Baker et al., 2016; Pástor & Veronesi, 2013) have provided evidence that uncertainty can significantly increase volatility in stock prices. An increase in stock volatility makes it difficult for investors to envisage stock prices and discourages them from trading (Hameed et al., 2010). Also, higher policy uncertainty decreases firm stock prices (Pástor & Veronesi, 2013). Duong et al. (2018) find that stock liquidity is reduced with increased uncertainty in policy.

In sum, we expect that uncertainty can weaken the FinTech and stock liquidity association. We measure uncertainty using the economic policy uncertainty (EPU) index developed by Baker et al. (2016) for the Chinese context.[4] We test the moderating role of EPU by introducing the interaction of EPU with all measures of FinTech (*FINTECH*EPU*, *FINTECH_BC*EPU*, and *FINTECH_DU*EPU*) in our main regression. Results presented in Table 13 indicate significantly positive coefficients on interaction terms in columns (1) to (6), whereas significantly negative coefficients in columns (7) to (12). These results provide support for our arguments that uncertainty (EPU) weakens the effect of FinTech on stock liquidity.

[Insert Table 13 here]

5. Conclusion

We show that the development of FinTech improves stock price liquidity. Specifically, firms operating in regions where FinTech is more developed have higher stock price liquidity. We argue that Fintech improves a firm's information environment by reducing information

[4] The details about this measure is discussed above.

asymmetry, leading to higher liquidity of stock prices. The findings are consistent with alternative measures of FinTech and stock liquidity. The results are also valid after addressing issues related to omitted variables and endogeneity. Our additional analyses suggest that the FinTech and stock liquidity association is stronger for firms with higher media coverage and analysts following. However, the association is weaker for SOEs and when firms face higher uncertainty.

Our findings provide several important implications for policymakers, regulators, and stock market participants. First, the results imply that FinTech development can significantly improve stock price liquidity. Hence it provides policy makers and regulators about the important role of FinTech development in enhancing the liquidity of stock prices in the Chinese context. FinTech can lead to efficient financial services for more corporations with demand, which effectively enhances the liquidity of stock prices. We should actively explore the deep application of FinTech at the corporate level, speed up the digital development of these firms, guide firms to apply digital finance, and further improve the deep evaluation of integrated development, to enhance the value of these firms. Therefore, listed firms should make reasonable use of FinTech to promote the information environment, to achieve the goal of improving corporate decisions. Third, this study offers implications for adopting FinTech that would significantly improve the information environment of firms' thereby providing benefits to firms. Greater liquidity would enhance investor participation in the capital market and play a significant role in its development leading to efficient utilization of resources. This is particularly true for emerging economies such as China, where the information environment is opaque. Finally, our findings also provide global implications, in particular to emerging economies, since our study is also focused on the emerging economy of China. These findings may help regulators and policy makers to know the important role of FinTech in stock price liquidity. Hence, policymakers in other countries must be aware that stock price liquidity can be enhanced through Fintech development. Because our analyses show that the positive effect of FinTech on stock liquidity is enhanced with higher media coverage and analysts following and weaker with higher uncertainty, these findings also provide implications. For example, policymakers and regulators should be aware of the important roles of media coverage, analysts, and uncertainty in the market as important factors that may affect the association of FinTech development and stock liquidity.

While our study provides several policy implications, it has also limitations. First, our study is based on the Chinese context, which unique institutional environment and FinTech development. Hence, the implications of the findings of the study are limited to the Chinese context. Nevertheless, it also provides an opportunity for exploring whether our conclusions hold in other economies. Second, while our analyses are based on the emerging markets context, considering China as the largest emerging economy, the conclusions are less likely to be applicable to the developed country's context. Third, our measure of FinTech proxy is based on the index developed by the Institute of Digital Finance of Peking University (PKU-digital finance IIC), the Shanghai Finance Institute, and Ant Financial Services Company. While this measured is the most valid, reliable, and widely used in literature, it is not a firm-level measure of the financial technology development of a company and it is possible that FinTech development varies across firms. However, this limitation also provides a future avenue for further development of financial technology at the firm level.

Reference

Aghanya, D., Agarwal, V., & Poshakwale, S. (2020). Market in Financial Instruments Directive (MiFID), stock price informativeness and liquidity. *Journal of Banking and Finance*, *113*, 105730.

Ali, S., Liu, B., & Su, J. J. (2017). Corporate governance and stock liquidity dimensions: Panel evidence from pure order-driven Australian market. *International Review of Economics and*

Finance, *50*, 275–304.
Allen, F., Qian, J., Zhang, C., & Zhao, M. (2012). *China's financial system: opportunities and challenges*. National Bureau of Economic Research.
Alph, S.O., Canbaloglu, B., & Gurgun, G. (2021). Stock liquidity, stock price crash risk, and foreign ownership. *Borsa Istanbul Review*, 22(3), 477 - 486.
Amihud, Y. (2002). Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, *5*(1), 31–56.
Amihud, Y., & Mendelson, H. (1986). Asset pricing and the bid-ask spread. *Journal of Financial Economics*, *17*(2), 223–249.
Ascioglu, A., Hegde, S. P., Krishnan, G. V., & McDermott, J. B. (2012). Earnings management and market liquidity. *Review of Quantitative Finance and Accounting*, *38*(2), 257–274.
Atawnah, N., Balachandran, B., Duong, H. N., & Podolski, E. J. (2018). Does exposure to foreign competition affect stock liquidity? Evidence from industry-level import data. *Journal of Financial Markets*, *39*, 44–67.
Aziz, A., & Naima, U. (2021). Rethinking digital financial inclusion: Evidence from Bangladesh. *Technology in Society*, *64*, 101509.
Baker, S. R., Bloom, N., Davis, S. J., Dashkeyev, V., Deriy, O., Dinh, E., Ezure, Y., Gong, R., Jindal, S., Kim, R., Klosin, S., Koh, J., Lajewski, P., Sachs, R., Shibata, I., Stephenson, C., & Takeda, N. (2016b). Measuring Economic Policy Unvertainity. *The Quarterly Journal of Economics*, *131*(4), 1593–1636.
Banna, H., & Alam, M. R. (2021). Impact of digital financial inclusion on ASEAN banking stability: implications for the post-Covid-19 era. *Studies in Economics and Finance*, *38*(2), 504–523.
Borisova, G., Brockman, P., Salas, J. M., & Zagorchev, A. (2012). Government ownership and corporate governance: Evidence from the EU. *Journal of Banking and Finance*, *36*(11), 2917–2934.
Boubaker, S., Gounopoulos, D., & Rjiba, H. (2019). Annual report readability and stock liquidity. *Financial Markets, Institutions and Instruments*, *28*(2), 159–186.
Boubakri, N., Chen, R. (Ryan), El Ghoul, S., Guedhami, O., & Nash, R. (2020). State ownership and stock liquidity: Evidence from privatization. *Journal of Corporate Finance*, *65*, 101763.
Boycko, M., Shleifer, A., & Vishny, R. W. (1994). Voucher Privitization. *Journal Financial Economimcs*, *35*, 249–266.
Brockman, P., Chung, D. Y., & Yan, X. S. (2009). Block Ownership , Trading Activity , and Market Liquidity. *The Journal of Financial and Quantitative Analysis*, *44*(6), 1403–1426.
Buchak, G., Matvos, G., Piskorski, T., & Seru, A. (2018). Fintech, regulatory arbitrage, and the rise of shadow banks. *Journal of Financial Economics*, *130*(3), 453–483.
Bushman, R. M. (2011). *Capital Allocation and Timely Accounting Recognition of Economic Losses*. *38*(October 2010), 1–33.
Butler, A. W., Grullon, G., & Weston, J. P. (2005). Can Managers Forecast Aggregate Market Returns? *Journal of Finance*, *60*(2), 963–986.
Cao, S., Nie, L., Sun, H., Sun, W., & Taghizadeh-Hesary, F. (2021). Digital finance, green technological innovation and energy-environmental performance: Evidence from China's regional economies. *Journal of Cleaner Production*, *327*, 129458.
Chatterjee, S., Hasan, I., John, K., & Yan, A. (2021). Stock liquidity, empire building, and valuation. Journal of Corporate Finance, 70, 102051.
Chen, S., & Zhang, H. (2021). Does digital finance promote manufacturing servitization: Micro

evidence from China. *International Review of Economics and Finance*, *76*, 856–869.
Chen, Z., Jin, J., & Li, M. (2022). Does media coverage influence firm green innovation？The moderating role of regional environment. *Technology in Society*, *70*, 102006.
Cheng, L., Xie, E., Fang, J., & Mei, N. (2022). Performance feedback and firms' relative strategic emphasis: The moderating effects of board independence and media coverage. *Journal of Business Research*, *139*, 218–231.
Cheng, M., & Qu, Y. (2020). Does bank FinTech reduce credit risk? Evidence from China. *Pacific Basin Finance Journal*, *63*, 101398.
Chung, K. H., Elder, J., & Kim, J. (2010). Corporate Governance and Liquidity. *Journal of Financial and Quantitative Analysis*, *45*(2), 265–291.
Conyon, M. J., & He, L. (2011). Executive compensation and corporate governance in China. *Journal of Corporate Finance*, *17*(4), 1158–1175.
Copeland, T. E., & Galai, D. (1983). Information Effects on the Bid-Ask Spread. *The Journal of Finance*, *38*(5), 1457–1469.
Corwin, S. A., & Schultz, P. (2012). A Simple Way to Estimate Bid-Ask Spreads from Daily High and Low Prices. *Journal of Finance*, *67*(2), 719–760.
Cumming, D., Meoli, M., & Vismara, S. (2021). Does equity crowdfunding democratize entrepreneurial finance? *Small Business Economics*, *56*(2), 533–552.
Dai, L., Shen, R., & Zhang, B. (2021). Does the media spotlight burn or spur innovation? *Review of Accounting Studies*, *26*(1), 343–390.
Dang, M., Puwanenthiren, P., Nguyen, M. T., Hoang, V. A., Mazur, M., & Henry, D. (2022). Does mangerial tone matter for stock liquidity? Evidence from textual disclosure. Finance Research Letter, 48, 102917.
Daud, S. N. M., Ahmad, A. H., Khalid, A., & Azman-Saini, W. N. W. (2022). FinTech and financial stability: Threat or opportunity? *Finance Research Letters*, 102667.
de Roure, C., Pelizzon, L., & Thakor, A. V. (2021). P2P Lenders versus Banks: Cream Skimming or Bottom Fishing? *Leibniz Institute for Financial Research SAFE.*
Demertzis, M., Merler, S., & Wolff, G. B. (2018). Capital markets union and the Fintech opportunity. *Journal of Financial Regulation*, *4*(2), 157–165.
Diamond, D. W., & Verrecchia, D. và (1991). Disclosure, liquidity, and the cost of capital. *Journal of Finance*, *46*(4), 1325–1359.
Ding, M., & Suardi, S. (2019). Government ownership and stock liquidity: Evidence from China. *Emerging Markets Review*, *40*, 100625.
Drake, S. M., Guest, M. N., & Twedt, J. B. (2014). The Media and Mispricing: The Role of the Business Press in the Pricing of Accounting Information. *Accounting Review*, *89*(5), 1673–1701.
Drechsler, I. (2013). Uncertainty, time-varying fear, and asset prices. *Journal of Finance*, *68*(5), 1843–1889.
Fang, V. W., Noe, T. H., & Tice, S. (2009). Stock market liquidity and firm value. *Journal of Financial Economics*, *94*(1), 150–169.
Fengrong, W., William, M., & Linda, M. (2022). Economic policy uncertainty and stock liquidity: The mitigating impact of informaion disclosure. Research in International Business and Finance, 59, 101553.
Fonseka, M., Samarakoon, L. P., Tian, G. L., & Seng, R. (2021). The impact of social trust and state ownership on investment efficiency of Chinese firms. *Journal of International Financial Markets, Institutions and Money*, *74*, 101394.

Frino, A., Palumbo, R., Capalbo, F., Gerace, D., & Mollica, V. (2013). Information disclosure and stock liquidity: Evidence from borsa Italiana. *Abacus*, *49*(4), 423–440.
Fuster, A., Plosser, M., Schnabl, P., & Vickery, J. (2019). The Role of Technology in Mortgage Lending. *Review of Financial Studies*, *32*(5), 1854–1899.
Gao, K., Lin, W., Yang, L., & Chan, K. C. (2020). The impact of analyst coverage and stock price synchronicity: Evidence from brokerage mergers and closures. *Finance Research Letters*, *33*, 101190.
Goldstein, I., Jiang, W., & Karolyi, G. A. (2019). To FinTech and beyond. *Review of Financial Studies*, *32*(5), 1647–1661.
Gomber, P., Koch, J. A., & Siering, M. (2017). Digital Finance and FinTech: current research and future research directions. *Journal of Business Economics*, *87*(5), 537–580.
Graf-Vlachy, L., Oliver, A. G., Banfield, R., König, A., & Bundy, J. (2020). Media Coverage of Firms: Background, Integration, and Directions for Future Research. *Journal of Management*, *46*(1), 36–69.
Hameed, A., Kang, W., & Viswanathan, S. (2010). Stock market declines and liquidity. *Journal of Finance*, *65*(1), 257–293.
Handa, P., & Schwartz, R. A. (1996). Limit order trading. *Journal of Finance*, *51*(5), 1835–1861.
Hasbrouck, J. (2006). Trading costs and returns for US equities: Estimating effective costs from …. *Unpublished Working Paper. New York University*, *LXIV*(3), 1445–1477. http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.121.6164&rep=rep1&type=pdf
Healy, P., & Palepu, K. (2001). Information asymmetry, corporate disclosure, and the capital markets: A review of the empirical disclostire literature. *Journal of Accounting and Economics*, *31*(1–3), 405–440.
Huang, R., Quan, L., Luo, H., Gao, Z., Lai, X., & Tang, S. (2024). Can FinTech promote enterprise cross-region investment?——Evidence from China. *International Review of Economics & Finance*, *96*, 103524.
Irani, R. M., & Oesch, D. (2016). Analyst Coverage and Real Earnings Management: Quasi-Experimental Evidence. *Journal of Financial and Quantitative Analysis*, *51*(2), 589–627.
Ji, Y., Shi, L., & Zhang, S. (2022). Digital finance and corporate bankruptcy risk: Evidence from China. *Pacific-Basin Finance Journal*, *72*, 101731.
Jiang, F., & Kim, K. A. (2015). Corporate governance in China : A modern perspective. *Journal of Corporate Finance*, *32*, 190–216.
Jiang, F., & Kim, K. A. (2020). Corporate Governance in China: A Survey. *Review of Finance*, *71432008*, 733–772.
Jiraporn, P., Chintrakarn, P., & Kim, Y. S. (2012). Analyst following, staggered boards, and managerial entrenchment. *Journal of Banking and Finance*, *36*(11), 3091–3100.
Jung, H., & Choi, S. (2021). The effects of employee stock ownership on stock liquidity: Evidence from the Korean market. *North American Journal of Economics and Finance*, *58*, 101468.
Kumari, J. (2019). Investor sentiment and stock market liquidity: Evidence from an emerging economy. *Journal of Behavioral and Experimental Finance*, *23*, 166–180.
Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, *53*(6), 1315–1336.
Lai, X., Yue, S., Guo, C., & Zhang, X. (2023). Does FinTech reduce corporate excess leverage? Evidence from China. *Economic Analysis and Policy*, *77*, 281-299.
Le, H., & Gregoriou, A. (2020). How Do You Capture Liquidity? a Review of the Literature on Low-Frequency Stock Liquidity. *Journal of Economic Surveys*, *34*(5), 1170–1186.

Levine, R., & Zervos, S. (1996). Stock market development and long-run growth. *World Bank Economic Review*, *10*(2), 323–339.
Liang, T., Zhang, Y. J., & Qiang, W. (2022). Does technological innovation benefit energy firms' environmental performance? The moderating effect of government subsidies and media coverage. *Technological Forecasting and Social Change*, *180*, 121728.
Liu, W. (2006). A liquidity-augmented capital asset pricing model. *Journal of Financial Economics*, *82*(3), 631–671.
Luo, S., Sun, Y., Yang, F., & Zhou, G. (2022). Does fintech innovation promote enterprise transformation? Evidence from China. *Technology in Society*, *68*, 101821.
Lv, P., & Xiong, H. (2022a). Can FinTech improve corporate investment efficiency? Evidence from China. *Research in International Business and Finance*, *60*, 101571.
Majeed, M. A., & Yan, C. (2022). Financial statement comparability and stock liquidity: evidence from China. *Applied Economics*, 1–18.
Masakara, Pakaj, K. (2020). The Role of P2P Platforms in Enhancing Financial Inclusion in US-An Analysis of Peer-to-Peer Lending Across the Rural-Urban Divide. *Available at SSRN 3715211.*, *23529*(2), 1–45.
Megginson, W. L. (2017). Privatization, State Capitalism, and State Ownership of Business in the 21st Century. *Foundations and Trends in Finance*, *11*(1–2), 1–153.
Michael, M., Ali, M. J., Atawnah, N., & Muniandy, B. (2022). Fiduciary or loyalty? Evidence from top management counsel and stock liquidity. *Global Finance Journal*, *52*, 100709.
Mollick, E., & Robb, A. (2016). Democratizing innovation and capital access: The role of crowdfunding. *California Management Review*, *58*(2), 72–87.
Moser, P., & Voena, A. (2012). Compulsory licensing: Evidence from the Trading with the Enemy Act. *American Economic Review*, *102*(1), 396–427.
Myers, S. C., & Majluf, N. S. (1984). Corporate financing and investment decisions when firms have information that investors do not have. *Journal of Financial Economics*, *13*(2), 187–221.
Nagar, V., Schoenfeld, J., & Wellman, L. (2019). The Effect of Economic Policy Uncertainty on Investor Information Asymmetry and Management Disclosures. *Journal of Accounting and Economics*, *67*(1), 36–57.
Pástor, Ľ., & Veronesi, P. (2013). Political uncertainty and risk premia. *Journal of Financial Economic Policy*, *110*(3), 520–545.
Pham, M. H. (2020). In law we trust: Lawyer CEOs and stock liquidity. *Journal of Financial Markets*, *50*, 100548.
Pollock, T. G., & Rindova, V. P. (2003). Media legitimation effects in the market for initial public offerings. *Academy of Management Journal*, *46*(5), 631–642.
Roy, P. P., Rao, S., & Zhu, M. (2022). Mandatory CSR expenditure and stock market liquidity. *Journal of Corporate Finance*, *72*, 102158.
Schoenfeld, J. (2017). The effect of voluntary disclosure on stock liquidity : New evidence from index funds. *Journal of Accounting and Economics*, *63*(1), 51–74.
Shao, J., Lou, Z., Wang, C., Mao, J., & Ye, A. (2022). The impact of artificial intelligence (AI) finance on financing constraints of non-SOE firms in emerging markets. *International Journal of Emerging Markets*, *17*(4), 930-944
Sheng, T., & Fan, C. (2020). Fintech, Optimal Banking Market Structure, and Credit Supply for SMEs. *Journal of Financial Research*, *480*(6), 114–132.
Tam, L., Dong, L., & Lam, K. (2019). Liquidity and Stock Returns: Evidence from the Chinese

Stock Market. *China Accounting and Finance Review*, *21*(4), 1–42.
Tang, S., Lai, X., Huang, R., (2019). How can FinTech innovation affect TFP: Facilitating or inhibiting? --Theoretical analysis framework and regional practice. *China Soft Science*, *7*, 134–144 (In Chinese).
Tang, X., Ding, S., Gao, X., & Zhao, T. (2022). Can digital finance help increase the value of strategic emerging enterprises? *Sustainable Cities and Society*, *81*, 103829.
Tang, M., Hou, Y. G., Goodell, J. W., & Hu, Y. (2024). Fintech and corporate risk-taking: Evidence from China. *Finance Research Letters*, *64*, 105411.
Tang, Z., & Tang, J. (2016). Can the Media Discipline Chinese Firms' Pollution Behaviors? The Mediating Effects of the Public and Government. *Journal of Management*, *42*(6), 1700–1722.
Trinh, H. H., Nguyen, C. P., Hao, W., & Wongchoti, U. (2021). Does stock liquidity affect bankruptcy risk? DID analysis from Vietnam. *Pacific Basin Finance Journal*, *69*, 101634.
Wang, K., & Zhang, X. (2021). The effect of media coverage on disciplining firms' pollution behaviors: Evidence from Chinese heavy polluting listed companies. *Journal of Cleaner Production*, *280*, 123035.
Wang, Q., Yang, J., Chiu, Y. ho, & Lin, T. Y. (2020). The impact of digital finance on financial efficiency. *Managerial and Decision Economics*, *41*(7), 1225–1236.
Wang, Q., Yang, L., & Yue, Z. (2022). Research on development of digital finance in improving efficiency of tourism resource allocation. *Resources, Environment and Sustainability*, *8*(September 2021), 100054.
Wang, X., & Wei, S. (2021). Does the investment horizon of institutional investors matter for stock liquidity? *International Review of Financial Analysis*, *74*, 101648.
Welker, M. (1995). Disclosure policy, information assymetry and liquidity in markets. *Contemporary Accounting Research*, *11*(2), 801–827.
Yu, M. (2010). Analyst forecast properties, analyst following and governance disclosures: A global perspective. *Journal of International Accounting, Auditing and Taxation*, *19*(1), 1–15.
Zhang, X., Wan, G., Zhang, J., He, Z., (2019). Digital Economy "Financial Inclusion" and Inclusive Growth. *Economic Research Journal*, 8, 71–86 (In Chinese).
Zhang, H., Luo, X., Han, M., & Liu, X. (2020). Cultural diversity in ownership and stock liquidity commonality: Evidence from China. *Finance Research Letters*, 101856.
Zhang, T., & Lence, H. S. (2022). Liquidity and asset pricing: Evidence from the Chinese stockmarkets. Norh American Journal of Economics and Finance, 59, 101557.
Zhao, J., Li, X., Yu, C. H., Chen, S., & Lee, C. C. (2022). Riding the FinTech innovation wave: FinTech, patents and bank performance. *Journal of International Money and Finance*, *122*, 102552.
Zhu, C. (2019). Big Data as a Governance Mechanism. *Review of Financial Studies*, *32*(5), 2021–2061.